

Mini EDA Project: Housing Data Analysis Report

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Introduction

This report provides a comprehensive exploratory data analysis (EDA) of a clean housing dataset comprising (545, 13) data i.e 545 rows and 13 columns. The dataset tracks key structural features such as property area, bedroom and bathroom counts, floor counts (stories), parking availability, and several categorical variables representing premium amenities (e.g., air conditioning, main road access, and furnishing status). The primary goal of this analysis is to identify statistical patterns and market trends that influence housing prices, ultimately providing data-driven recommendations to guide a hypothetical value-conscious homebuyer.

Key Findings and Insights

- **Market Skew and Price Benchmarks:** The average property price in the dataset sits at approximately **\$4,766,729.25**, whereas the median price is lower at **\$4,340,000.00**. This structural variance indicates a right-tailed distribution skewed by high-value, luxury property outliers, which peak at a maximum price of **\$13,300,000.00**.
- **The Spatial Premium:** There is a moderately strong positive correlation of **0.54** between a property's total area and its final selling price. This indicates that while larger spatial dimensions reliably command higher market rates, area size alone does not dictate property value.
- **Characteristics of Premium Real Estate:** An evaluation of the five most expensive homes reveals a strong preference for premium infrastructure. All five feature direct access to a main road, four out of five are fully furnished, and four out of five are equipped with central air conditioning systems.
- **Structural Composition Trends:** The average home in this market is optimized for mid-sized family occupancy, presenting a mean bedroom count of **2.97** and an average of **1.29** bathrooms. Properties generally feature close to two stories (a mean of **1.81**) and average **0.69** dedicated parking spaces.

Visualizations

To analyze the underlying market distributions and spatial impacts on valuation, the following charts were plotted during the EDA process:

1. Housing Price Distribution

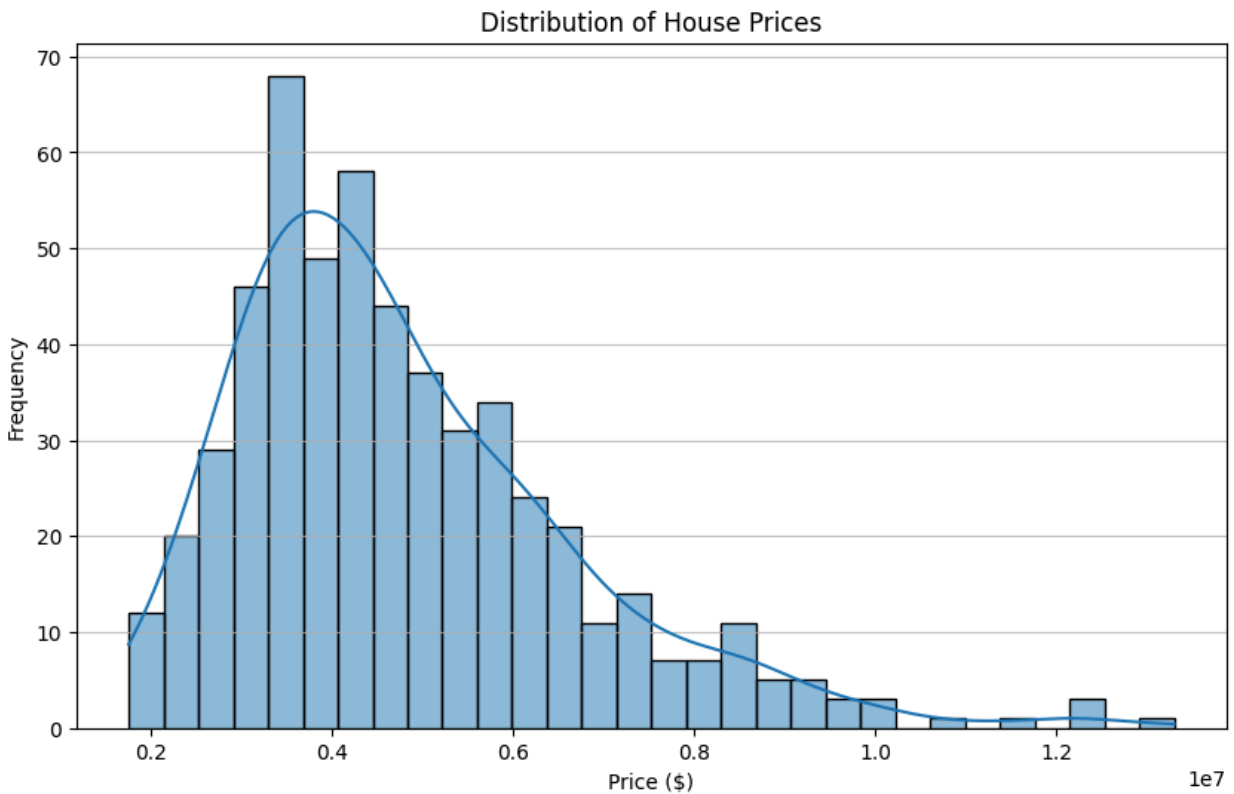


Figure 1: This plot illustrates a right-skewed distribution where most home entries cluster between \$3.0M and \$6.0M, highlighting a select tier of ultra-luxury properties stretching past \$10.0M.

2. Price vs. Property Area

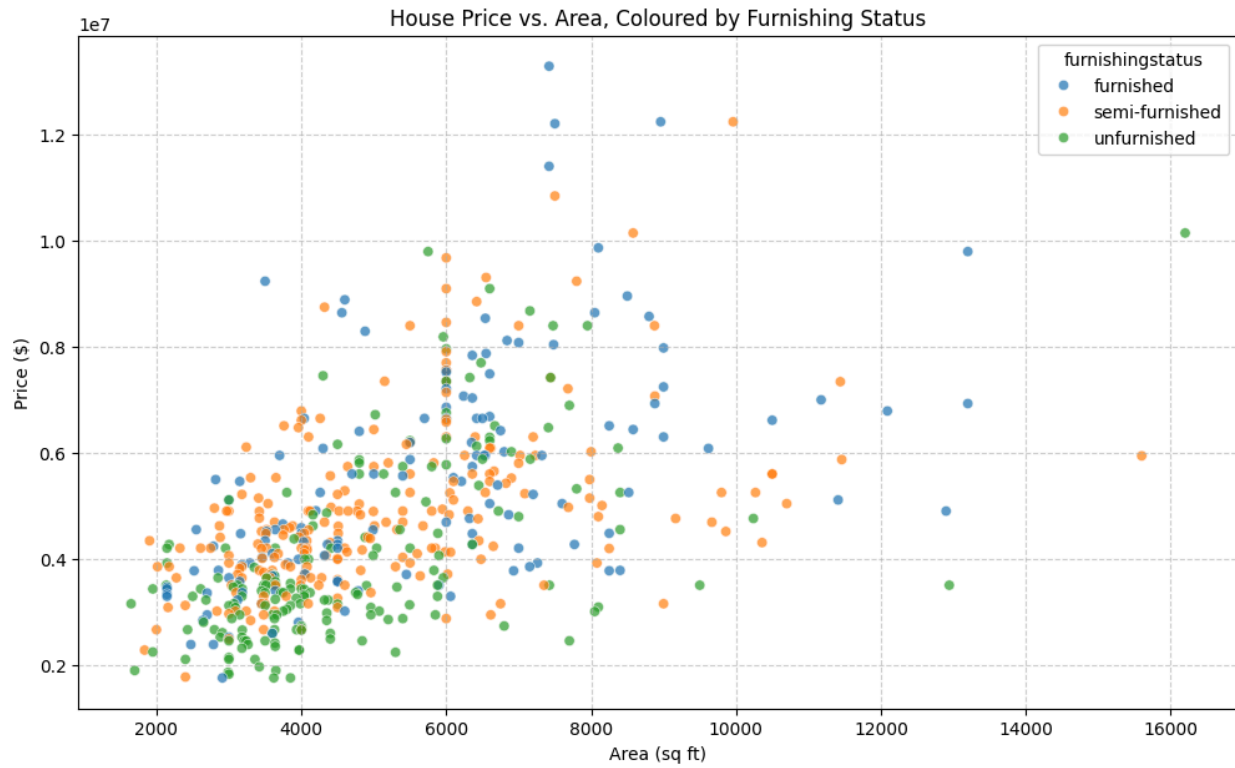


Figure 2: The scatter profile displays the 0.54 correlation baseline. Upward price dispersion widens significantly past the median area mark of 4,600 sq ft, indicating that secondary luxury features compound the pricing of larger homes.

3. House Pricing By Air Conditioning Status

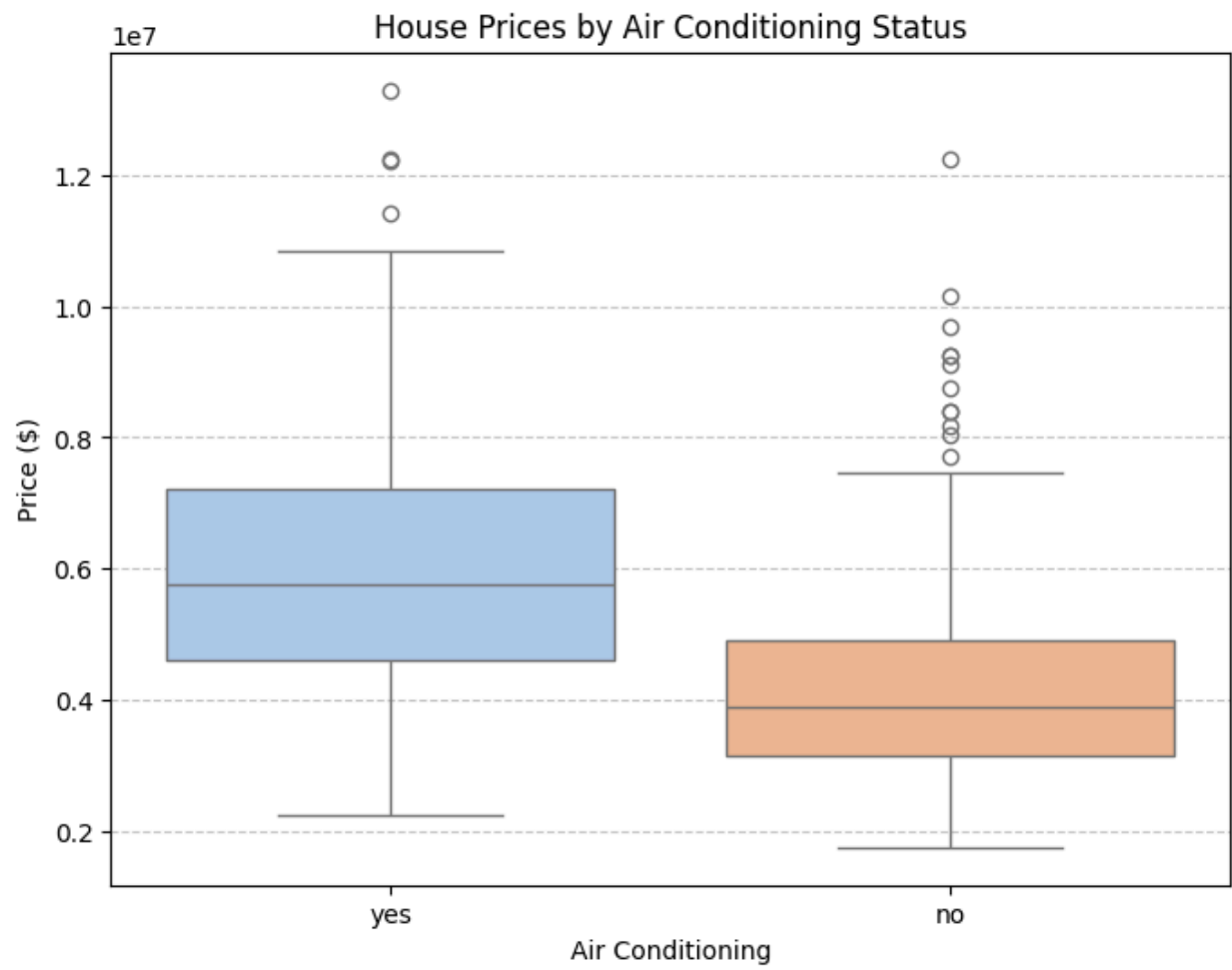


Figure 3: Houses with AC have a higher median price (~\$5.75M) than those without (~\$3.85M), with a wider price spread and several high-end outliers in both groups.

4. House Pricing By Number of bedroom



Figure 4: Price rises with bedroom count up to 5 (\$5.8M), then dips at 6 (\$4.8M) — likely a small-sample effect, worth checking row counts.

5. Correlation Matrix

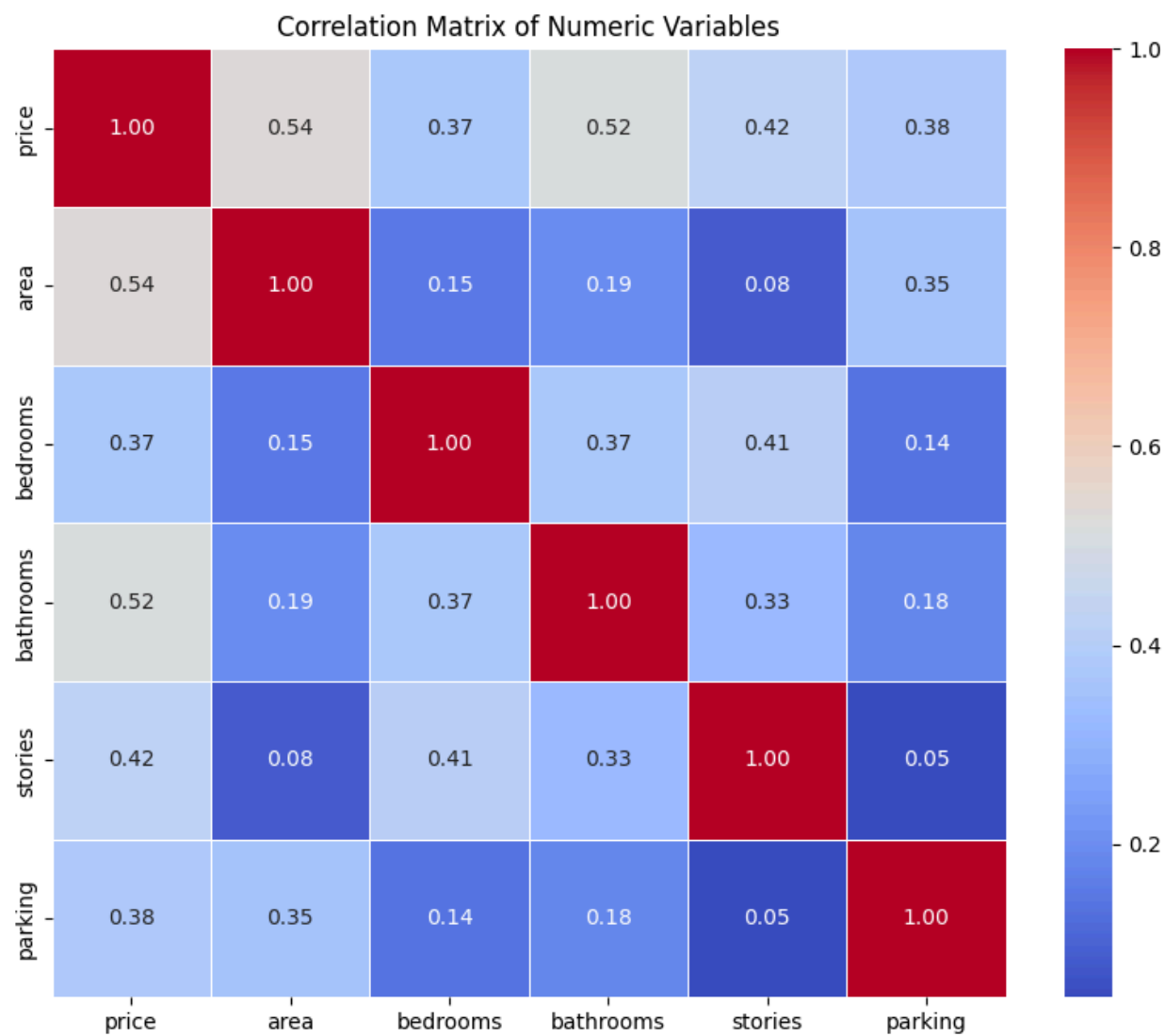


Figure 5: Price correlates most with area (0.54) and bathrooms (0.52), then stories (0.42), parking (0.38), and bedrooms (0.37) — no single feature dominates.

Conclusion and Strategic Recommendations

The exploratory analysis confirms that while spatial area establishes a reliable baseline price floor, premium categorical traits (such as modern air conditioning, main road positioning, and structural tiering) drive top-tier market valuations.

For a hypothetical house buyer looking to maximize asset value, the following strategies are recommended:

1. **Target the Median Cluster:** Focus search parameters on homes priced near the median (\$4.34M) rather than the arithmetic average (\$4.77M). This range bypasses the luxury premium inflation driven by extreme market outliers.
2. **Optimize the Area-to-Amenity Ratio:** Since area has a moderate correlation of 0.54 with price, buyers should look for homes with slightly smaller footprints (~4,000–4,500 sq ft) situated in good locations. This captures functional utility without triggering the exponential price curve seen in sprawling plots.
3. **Leverage Unfurnished Arbitrage:** Because the most expensive properties show a high concentration of fully furnished setups, value seekers should prioritize "unfurnished" or "semi-furnished" options. Furnished homes average \$5.50M vs unfurnished \$4.01M is a ~\$1.49M gap that buyers can capture. These properties carry a lower upfront market cost, allowing buyers to save capital and add equity through independent renovations over time.